Hybrid Voltage Control in Distribution Networks Under Limited Communication Rates

Hao Jan Liu, *Member, IEEE*, Wei Shi, *Member, IEEE*, and Hao Zhu, *Member, IEEE*

***Abstract*— Voltage regulation in distribution networks is challenged by increasing penetration of distributed energy resources (DERs). Thanks to advancement in power electronics, DERs can be leveraged to regulate the grid voltage by quickly changing their reactive power outputs. This paper develops a hybrid voltage control (HVC) strategy that can seamlessly integrate both local and distributed designs to effectively coordinate the network-wide DERs. By minimizing a special voltage mismatch objective, we achieve the proposed HVC design by adopting partial primal-dual gradient updates that can allow for a distributed and online implementation. The proposed HVC design improves over existing distributed approaches by reliably integrating local voltage feedback. As a result, it can dynamically adapt to varying system operating conditions while being fully cognizant of the instantaneous availability of communication links. Under the worst-case scenario of total link outages, the proposed design naturally boils down to a surrogate local control implementation. Numerical tests on realistic feeder cases have been made to corroborate our analytical results and demonstrate the algorithmic performance.**

***Index Terms*— Volt/VAR optimization (VVO), voltage regulation, reactive power control, distributed optimization, distribution systems, partial primal-dual algorithm.**

I. Introduction

RECENT proliferation of distributed energy resources (DERs) such as solar generation and storage devices can potentially cause some rapid voltage fluctuations in distribution networks. A promising approach to tackle this challenge is via advanced inverter control design. The fast-acting power electronic-interfaced DERs can support the distribution system voltage regulation objective by managing their *reactive power* (VAR) outputs in a coordinated fashion.

The voltage control problem can be viewed as a special case of the optimal power flow (OPF) one that instead aims to minimize the voltage mismatch. Under this formulation, a centralized approach would require the availability of full network-wide information [1]. It is also possible to develop local control strategies based on bus voltage magnitude information [2], [3]. Our earlier work [4], [5] has also provided the performance and stability analysis for local designs under both static and dynamic operating conditions, and has identified their fundamental limitations due to lack of real-time information exchange. At affordable communication complexity, several distributed optimization based techniques using neighborhood information exchanges have been proposed to achieve the global optimum [6]–[10]; see [11] for a tutorial review of recent distributed and decentralized methods. Albeit a distributed control improves the voltage mismatch over a local one, the performance of the former highly depends on the availability and quality of communication links, in particular the rate of information exchange as pointed out by [11]. While the effects of bandwidth limits and messages quantization have been investigated in [12] for distributed voltage control, more focus has been put on the more practical scenario of low-rate communications, or equivalently, asynchronous control updates. For example, [13] has designed an asynchronous decentralized algorithm where each DER controller flexibly integrates the fast incoming local voltage information with the low-rate control signal from the centralized aggregator. Also, we have investigated the robustness of the fully distributed and online design against random failures of bus-to-bus communication links in [10]. Nonetheless, this fully distributed approach would fail to incorporate any local voltage information under the worst-case scenario of total communication outages. This is because each node needs to freeze all variable updates in order to stay coordinated with neighboring nodes. Hence, to reduce communication complexity and enhance robustness to imperfect communications, it is imperative to develop an integrated design that can achieve the dual objectives in terms of flexible adaptivity to variable rate of communications and global optimality of voltage regulation performance. Such an innovative design has the potential of unifying the currently separated framework of either local or distributed control design of [4], [5], and [10].

The present paper aims to design a hybrid voltage control (HVC) strategy that can dynamically adapt to varying system operating conditions while being fully cognizant of the instantaneous rates of communication links. To cope with practical communication limitations, the proposed HVC scheme consists of both distributed and local control designs and does not require a centralized authority. Different from the traditional hierarchical control architecture including the primary

Manuscript received July 21, 2017; revised October 25, 2017 and December 13, 2017; accepted January 14, 2018. Date of publication January 24, 2018; date of current version April 19, 2019. This work was supported in part by the National Science Foundation under Award ECCS-1610732, and in part by the Power Systems Research Engineering Center. Paper no. TSG-01027-2017. *(Corresponding author: Hao Jan Liu.)*

H. J. Liu is with the Information Trust Institute, University of Illinois at Urbana–Champaign, Urbana, IL 61801 USA (e-mail: haoliu6@illinois.edu).

W. Shi is with the Department of Electrical and Computer Engineering, Boston University, Boston, MA 02215 USA (e-mail: wilburs@bu.edu).

H. Zhu is with the Department of Electrical and Computer Engineering, University of Texas at Austin, Austin, TX 78712 USA (e-mail: haozhu@utexas.edu).

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Digital Object Identifier 10.1109/TSG.2018.2797692

and secondary layers, the strength of our control design lies in a seamlessly integrated framework with guaranteed performance without assuming a time-scale separation. We formulate the network-wide VAR optimization problem based on the linearized power flow model for analysis purposes only. The resultant quadratic programming problem is solved by a *partial* primal-dual gradient (PPD) algorithm in discrete-time domain, which is a variant of the classical primal-dual (sub)gradient method; see, [14]. We provide the analysis in step-size choices that can guarantee convergence. We further use the PPD-based solver to design online HVC strategy where each bus can integrate both the local voltage measurement and the communication information shared by neighboring buses. Although a linearized model has been adopted to the algorithmic development and analysis, performance of the proposed HVC design has been verified using the full ac power flow model for unbalanced and lossy distribution networks.

Compared to existing voltage control approaches, the main contributions of our HVC design are three-fold. First, it explicitly accounts for the VAR limits by using the projection operator. Due to the discontinuity of projection mapping, general Krasovskii's methods [15] for analyzing the stability of primal-dual gradient flow method would not hold. To tackle this problem, we have expressed the operation as a subgradient step featured by an indicator function in order to establish the stability of the PPD-based HVC design in Section III-A. Second, the HVC design only requires each bus to measure its local voltage magnitude and communicate it to neighboring buses. The sensing requirement and communication overhead are minimal in addition to simple control algorithmics compared to most (de)centralized strategies. Last but not least, our hybrid design can integrate both the neighboring bus voltage information and local voltage measurements regardless of the communication link conditions. This way, the HVC design not only dynamically but also automatically adapts to time-varying operating conditions. Meanwhile, it is cognizant of the instantaneous availability of communication links and effectively tracks the globally optimal VAR setting. Interestingly, although the HVC updates have been developed using the distributed PPD-based solver, it would boil down to a surrogate local voltage control update during a total communication outage. Under this worst-case scenario, satisfactory performance can still be achieved as it responds to local voltage variations.

This paper is organized as follows. Section II presents the power flow model for distribution networks and formulates the voltage mismatch minimization problem. Section III develops the PPD-based HVC algorithm at guaranteed optimality, along with its online implementation and communication-cognizant design. Numerical test results using realistic feeders and real-time-series data are presented in Section IV, with the paper wrapped up in Section V.

II. System Modeling and Problem Statement

Consider first a single-phase distribution network given by a graph with the set of buses $\mathcal{N} := \{0, \dots, N\}$ and the set of line segments $\mathcal{E} := \{(i, j)\}$, where the head of the feeder (substation or secondary-side of voltage regulator) is denoted by bus 0. Note that typical voltage regulation devices such as on-load tap changers (OLTCs) have a much slower dynamics compared to power electronic interfaced DERs. Under this time-scale separation, slowly acting devices are inactive and thus ignored for the fast inverter-based VAR control problem, as commonly assumed by earlier work; see [4], [6], [8], [13] and references therein. Per bus j, let v_j denote its voltage magnitude, and p_j (q_j) represent the active (reactive) power injection, respectively. All network quantities are in per unit (p.u.). A reference bus voltage v_0 is assumed constant. For each line (i, j), we denote r_{ij} and x_{ij} as its resistance and reactance in addition to P_{ij} and Q_{ij} as the active and reactive power flow from i to j, respectively.

The so-termed LinDistFlow model in [16] is adopted to linearize the distribution flow model, assuming negligible line losses and almost flat voltage. The accuracy of LinDistFlow and its multi-phase unbalanced generalization in [17] has been corroborated by several recent work [4], [6], [8], [10], [18]. Per bus j, the controllable VAR from inverters $q_j^{\mathrm{g}} := q_j + q_j^{\mathrm{c}}$ is the decision variable for given VAR consumption q_j^{c}. Using lineraization, the voltage v_j can be related to q_j^{g} as [4]

$$\sum_{i \in \mathcal{N}_j} B_{ji} v_i = q_j^{\mathrm{g}} + w_j, \ \forall j \in \mathcal{N} \tag{1}$$

where $\mathcal{N}_j := \{i | (i, j) \in \mathcal{E}\} \cup \{j\} \subseteq \mathcal{N}$ contains bus j and all of its neighboring buses. The quantity w_j captures the system operating conditions; i.e., the aggregated effect of $\{p_j, \ q_j^{\mathrm{c}}\}$ everywhere when $q_j^{\mathrm{g}} = 0, \ \forall j$. By concatenating all scalar variables into vectors and replacing $\mathbf{q}^{\mathrm{g}}$ by $\mathbf{q}$ for notational convenience, one can represent (1) in the compact form $\mathbf{B}\mathbf{v} = \mathbf{q} + \mathbf{w}$. Matrix $\mathbf{B}$ is the *Bbus matrix* as in the dc power flow model (see [19, Sec. 6.16]). By defining its inverse $\mathbf{X} = \mathbf{B}^{-1}$, one can also write the power flow model as $\mathbf{v} = \mathbf{X}(\mathbf{q} + \mathbf{w})$. By definition, matrix $\mathbf{B}$ is a reduced, weighted graph Laplacian matrix (full rank) with a unique sparsity pattern based on the network topology; i.e., $B_{ij} = B_{ji} = 0, \forall (i, j) \notin \mathcal{E}$. The extension of $\mathbf{B}$ to multi-phase unbalanced networks is possible and maintains the topology-based sparsity structure [17], [20]. Thus, the linearized model (1) and our proposed algorithm are readily applicable for realistic feeder systems as shown by numerical tests, while the presentation will focus on the simpler single-phase systems.

The goal is to optimize $\mathbf{q}$ contributed from network-wide inverters that can achieve a desired voltage profile $\boldsymbol{\mu}$ such that $\mathbf{v} \rightarrow \boldsymbol{\mu}$. This voltage regulation objective is related to the design of *secondary* voltage control in transmission and microgrid systems, by coordinating network-wide VAR resources to provide better voltage support during external disturbances; see [21], [22]. More general operational objectives can be obtained too; see Remark 1 for more discussions. We start with the static problem formulation where the operating condition $\mathbf{w}$ stays constant. The HVC algorithm design will be eventually extended to be fully online and distributed where each bus j only needs the neighboring voltage information to dynamically track w_j. The voltage mismatch error objective $f_1(\mathbf{v}) := \frac{1}{2}\|\mathbf{v} - \boldsymbol{\mu}\|^2$ is introduced to provide the *globally* optimal VAR setting. Minimizing f_1 subject to (1) could be solved collaboratively by the buses using distributed optimization techniques; see [6], [8]–[10]. Nonetheless, these distributed designs would fail to adapt to a total communication outage, during which isolated buses have to freeze all local variables in order to remain coordinated with neighboring buses. To make the control design more robust, we introduce also a *weighted* error objective $f_2(\mathbf{v}) := \frac{1}{2}\|\mathbf{v} - \boldsymbol{\mu}\|^2_{\mathbf{B}}$ where $\|\mathbf{y}\|^2_{\mathbf{B}} := \mathbf{y}^\top \mathbf{B}\mathbf{y}$ for any vector $\mathbf{y}$. It will turn out that the gradient direction for minimizing f_2 only depends on the local voltage deviation, as in the droop design [4]. Under limited VAR resources, this weighted objective f_2 is not equivalent to the original one f_1. However, incorporating the former turns out to be extremely useful for achieving the HVC design that can seamlessly perform local updates even at no incoming messages from neighboring.

Towards achieving the attractive features of both distributed and local control designs, we cast the HVC problem as

$$(\mathbf{v}^\star, \mathbf{q}^\star) := \arg\min_{\mathbf{v},\mathbf{q}} \quad \left[f_1(\mathbf{v}) + \gamma f_2(s(\mathbf{q}))\right] \tag{2a}$$

$$\text{subject to} \quad \mathbf{B}\mathbf{v} = \mathbf{q} + \mathbf{w} \tag{2b}$$

$$\underline{\mathbf{q}} \le \mathbf{q} \le \overline{\mathbf{q}} \tag{2c}$$

where $s(\mathbf{q}) := \mathbf{X}(\mathbf{q} + \mathbf{w}) = \mathbf{v}$ according to (1). Both $\mathbf{v}$ and $\mathbf{q}$ are the decision variables in the problem (2), with either one uniquely determined by the other. However, expressing f_2 as a function of $\mathbf{q}$ would allow for conveniently forming its instantaneous gradient direction using local voltage mismatch error. Note that the problem (2) is strongly convex because $\mathbf{B}$ is positive definite [4], and thus the optimum $(\mathbf{v}^\star, \mathbf{q}^\star)$ is unique. The objective function (2a) uses a positive parameter $\gamma > 0$ to balance between f_1 and f_2, depending on the communication rates. Under very frequent link failures, γ needs to increase to weigh more on f_2 and thus the local voltage error information. If high-rate communications are available, a small γ value would be preferred to minimize the global voltage mismatch error f_1. Hence, the choice of γ can be determined upon assessing the quality of communication network in order to balance between the global voltage mismatch error and the local one. Furthermore, the sparsity structure of $\mathbf{B}$ based on the network topology would induce coupling only among the neighborhood $\mathcal{N}_j$ in (2b). This is instrumental for the development of a distributed HVC solution. Moreover, the bounds on each local VAR resource in (2c) are constrained by inverters' apparent power limit and/or power factor limit, and thus they depend on the instantaneous active power outputs. If there are abundant VAR resources, (2c) would become inactive. Under this scenario, the optimal VAR output $\mathbf{q}^\star$ is the same regardless of the choice of γ, as $\mathbf{v}^\star = \boldsymbol{\mu}$ can always be achieved. This fact justifies the inclusion of f_2 as its solution could closely approximate the globally optimal one to f_1. Last, the modeling of inverter controllers is not needed for our system-level problem [8], [9], [18]. Thanks to the time-scale separation between the dynamics of internal inverter control and distribution system level disturbances [23], the former has a very minimal effect on the network VAR control problem [24].

Remark 1 (General Operational Objectives): The network-wide voltage regulation objective follows from the secondary voltage control function of transmission and microgrid systems; see [21], [22]. One can achieve various operational tasks or specifications by designing the preferred voltage profile $\boldsymbol{\mu}$. For high voltage quality or effective conservation voltage reduction [25], a flat voltage profile $\boldsymbol{\mu} = \mathbf{1}$ would be preferred. Moreover, the distribution management system (DMS) can perform network-specific planning studies to determine the best $\boldsymbol{\mu}$, as in earlier work [4], [6], [10], [20]. For example, one can choose a decreasing voltage profile to potentially reduce system losses, compensate voltage unbalance, or correct power factor. Nonetheless, the selection of $\boldsymbol{\mu}$ would be infrequently performed and not affect the implementation of our proposed HVC algorithm. Last but not least, the quadratic error objective f_1 can also be generalized to more robust error criterion such as the Huber loss function; see [26, Ch. 7]. Remark 2 will show that the performance analysis also holds for more generalized error objective.

III. Hybrid Voltage Control

This section presents our proposed hybrid voltage control (HVC) framework. Aiming to achieve the features of both distributed and local designs, we solve (2) by adopting a specific type of primal-dual gradient algorithm. Many variants of this algorithm have been studied in both continuous-time domain (see [27], [28]) and discrete-time domain (see [14], [29]). Under our result HVC design, each bus needs only to measure and exchange its own voltage magnitude information. As detailed soon, this feedback based approach differs from most distributed optimization schemes for power systems that have been developed under the static setup, with limited considerations of online implementations and imperfect communications. Interestingly, our HVC scheme would boil down to a surrogate local voltage control update under the worst-case scenario of a total communication outage. Thus, it enjoys satisfactory performance guarantees while being cognizant of varying communication rates.

To this end, with multiplier $\boldsymbol{\lambda}$ introduced for (2b), we obtain the following Lagrangian function for the static problem (2):

$$\mathcal{L}(\mathbf{v}, \mathbf{q}, \boldsymbol{\lambda})_{\mathbf{q}\in\mathcal{Q}} = \frac{1}{2}\|\mathbf{v} - \boldsymbol{\mu}\|^2 + \frac{\gamma}{2}\|\mathbf{X}(\mathbf{q} + \mathbf{w}) - \boldsymbol{\mu}\|^2_{\mathbf{B}} + \langle \boldsymbol{\lambda}, \mathbf{B}\mathbf{v} - \mathbf{q} - \mathbf{w}\rangle \tag{3}$$

where the set $\mathcal{Q} := \{\mathbf{q} | \mathbf{q} \in [\underline{\mathbf{q}}, \overline{\mathbf{q}}]\}$ and $\langle\cdot,\cdot\rangle$ represents the inner product. The problem now becomes to seek the saddle point of (3), which contains the global optimum of (2) under strong convexity. One popular and efficient method towards this goal is to instead construct the augmented Lagrangian by the quadratic norm of constraint violation and perform alternating minimization over primal variables followed by a (sub)gradient ascent update over dual variables (see [30] for a tutorial overview of ADMM). Unfortunately, the quadratic term in the augmented Lagrangian would result in computation tasks either at a centralized location or over the multi-hop neighborhood. This connectivity requirement is more challenging to implement while less robust to link failures. Hence, we focus on the Lagrangian function (3) and adopt the iterative *partial* primal-dual (PPD) method. The resultant solution admits to a sparse communication network in accordance to the distribution network topology while being able to cope up with variable communication rates. The "partial" update property will become more clear after presenting the algorithmic details.

Using the superscript to denote the iteration index k, we initialize $\boldsymbol{\lambda}^0 = \mathbf{0}$ and $\mathbf{q}^0$ to be the latest VAR setting. The PPD updates at the $(k+1)$-st iteration consists of the following three steps:

(S1) Update $\mathbf{v}$: For given $\mathbf{q}^k$ and $\boldsymbol{\lambda}^k$, $\mathbf{v}$ is updated by solving $\mathbf{v}^{k+1} = \arg\min_{\mathbf{v}} \mathcal{L}(\mathbf{v}, \mathbf{q}^k, \boldsymbol{\lambda}^k)$, which is an unconstrained quadratic program, for which there exists closed-form solution. Note that the Lagrangian $\mathcal{L}$ in (3) is separable over v_j's, and thus the update decouples into each bus j as

$$v_j^{k+1} = -\sum_{i \in \mathcal{N}_j} B_{ji}\lambda_i^k + \mu_j. \tag{4}$$

The iterates $\mathbf{v}^k$ can be viewed as an estimate of the network voltage based on the power flow model (1) This $\mathbf{v}$ update via direct minimization is different from the classical gradient flow algorithm, which would resort to a gradient descent typed update like $\dot{\mathbf{v}} = -\frac{\partial}{\partial \mathbf{v}}\mathcal{L}(\mathbf{v}, \mathbf{q}^k, \boldsymbol{\lambda}^k)$. Hence, the term "*partial*" is used here to acknowledge this difference. Similar strategy of "partial gradient update" has been explored in [28] for a continuous-time setting whereas our focus is on the discrete-time update.

(S2) Update $\mathbf{q}$: Using the step-size $\alpha > 0$, we perform a gradient-projection based update on $\mathbf{q}$, as given by

$$\mathbf{q}^{k+1} := \left\{\mathbf{q}^k - \alpha \nabla_{\mathbf{q}}\mathcal{L}\left(\mathbf{v}^{k+1}, \mathbf{q}^k, \boldsymbol{\lambda}^k\right)\right\} \tag{5}$$

where the operator $\mathcal{P}\{\cdot\}$ projects any input into the box $\mathcal{Q}$. Hence, $\mathbf{q}^k \in [\underline{\mathbf{q}}, \overline{\mathbf{q}}]$ always holds at any time k, and this iterate serves as the feasible control signal to command the VAR resources. Under this setting and using $\mathbf{XB} = \mathbf{I}$, the gradient direction in (5) becomes

$$\begin{aligned}\nabla_{\mathbf{q}}\mathcal{L}\left(\mathbf{v}^{k+1}, \mathbf{q}^k, \boldsymbol{\lambda}^k\right) &= \gamma \mathbf{XB}\left[\mathbf{X}\left(\mathbf{q}^k + \mathbf{w}\right) - \boldsymbol{\mu}\right)\Big] - \boldsymbol{\lambda}^k \\ &\approx \gamma\left(\tilde{\mathbf{v}}^k - \boldsymbol{\mu}\right) - \boldsymbol{\lambda}^k\end{aligned} \tag{6}$$

where the actual system voltage $\tilde{\mathbf{v}}^k \approx \mathbf{X}(\mathbf{q}^k + \mathbf{w})$ if the VAR outputs equals to $\mathbf{q}^k$, following from the linearized model (1). Note that each $\tilde{v}_j^k$ can be easily measured per bus j. Thus, computing the gradient direction in (6) completely decouples per bus j upon measuring its voltage $\tilde{v}_j^k$ and forming its multiplier λ_j^k. As both matrices $\mathbf{X}$ and $\mathbf{B}$ are uniquely invertible of each other, any weighting matrices other than $\mathbf{B}$ will not lead to this decoupling feature. And this is exactly the reason for choosing f_2 in the objective function. Thanks to the separability of box constraints, we further decouple the update (5) into each bus j as

$$q_j^{k+1} := \mathcal{P}_j\left\{q_j^k - \alpha\left[\gamma\left(\tilde{v}_j^k - \mu_j\right) - \lambda_j^k\right]\right\} \tag{7}$$

where $\mathcal{P}_j$ denotes the projection at bus j to $[\underline{q}_j, \overline{q}_j]$. Even though these bounds would vary under dynamic setting based on instantaneous active power outputs of inverters, they can be easily updated at each bus. Note that the gradient update in (7) using the voltage measurement also nicely adapts to dynamic setting, as the physical power flow coupling always guarantees it has the latest network-wide information. This unique feature will enable the robustness of our proposed HVC design against link failures. Thus, the iterate q_j^{k+1} can be considered as an implementable *transient solution* that is actively tracking the dynamic optimality under time-varying operating conditions.

(S3) Update $\boldsymbol{\lambda}$: For a given step-size $\beta > 0$, each multiplier per bus j is linearly updated using the iterative mismatch of the respective equality constraint, as given by

$$\lambda_j^{k+1} := \lambda_j^k + \beta\left[\sum_{i \in \mathcal{N}_j} B_{ji}v_i^{k+1} - q_j^{k+1} - w_j\right]. \tag{8}$$

The PPD-based iterations in (S1)-(S3) constitute the basis for our proposed HVC design. As mentioned earlier, for multi-phase networks, the generalized matrix $\mathbf{B}$ would have a *block* sparse structure [17]. Hence, it suffices to replace each scalar variable in the single-phase formulation by its three-phase vector counterpart for the multi-phase extensions of (S1)-(S3).

A. Convergence Analysis

With no projection operation in the $\mathbf{q}$-update (5), the convergence problem of PPD boils down to a straightforward stability analysis of a discrete-time linear system. However, the presence of projection is equivalent to the so-called saturation effects in linear systems, for which the stability analysis could be much more challenging [31]. To tackle this problem, we instead adopt a convex analysis approach.

To streamline the presentation, consider the following general form of (3):

$$\mathcal{L}(\mathbf{v}, \mathbf{q}, \boldsymbol{\lambda}) = f(\mathbf{v}) + g(\mathbf{q}) + \mathcal{I}(\mathbf{q}) + \langle \boldsymbol{\lambda}, \mathbf{Bv} - \mathbf{q} - \mathbf{w}\rangle \tag{9}$$

where both functions f and g are strongly convex and have Lipschitz gradients. To account for $\mathcal{Q}$, define the following indicator function

$$\mathcal{I}(\mathbf{q}) = \begin{cases} 0, & \text{if } \mathbf{q} \in \mathcal{Q}, \\ +\infty, & \text{if } \mathbf{q} \notin \mathcal{Q}. \end{cases}$$

Thus, (3) is a special case of (9) with $f(\mathbf{v}) \triangleq f_1(\mathbf{v}) = \frac{1}{2}\|\mathbf{v}-\boldsymbol{\mu}\|^2$ and $g(\mathbf{q}) \triangleq f_2(s(\mathbf{q})) = \frac{\gamma}{2}\|\mathbf{X}(\mathbf{q}+\mathbf{w}) - \boldsymbol{\mu}\|_{\mathbf{B}}^2$. Thus, there exist positive constants η and L such that $g(\cdot)$ is a η-strongly convex function with L-Lipschitz gradient; i.e., it holds that

$$\begin{aligned}\langle \mathbf{x} - \mathbf{y}, \nabla g(\mathbf{x}) - \nabla g(\mathbf{y})\rangle &\geq \eta\|\mathbf{x} - \mathbf{y}\|_2^2, \ \forall \mathbf{x}, \mathbf{y} \\ \|\nabla g(\mathbf{x}) - \nabla g(\mathbf{y})\|_2 &\leq L\|\mathbf{x} - \mathbf{y}\|_2, \ \forall \mathbf{x}, \mathbf{y}.\end{aligned}$$

In addition, $f(\cdot)$ is c-strongly convex for any $c = 1$ while $\mathcal{I}(\cdot)$ is a convex function.

The saddle point $(\mathbf{v}^\star, \mathbf{q}^\star, \boldsymbol{\lambda}^\star)$ to (9) containing the optimal solution $(\mathbf{v}^\star, \mathbf{q}^\star)$ to (2) satisfies the following KKT conditions:

$$\nabla f\big(\mathbf{v}^\star\big) + \mathbf{B}^\top \boldsymbol{\lambda}^\star = \mathbf{0}, \tag{10a}$$

$$\nabla g\big(\mathbf{q}^\star\big) - \boldsymbol{\lambda}^\star + \partial \mathcal{I}\big(\mathbf{q}^\star\big) \ni \mathbf{0}, \tag{10b}$$

$$\mathbf{B}\mathbf{v}^\star - \mathbf{q}^\star - \mathbf{w} = \mathbf{0} \tag{10c}$$

where $\partial\mathcal{I}(\mathbf{q}^\star)$ is the subdifferential set of $\mathcal{I}(\cdot)$ at $\mathbf{q}^\star$, which contains any subgradient of $\mathcal{I}(\cdot)$ at $\mathbf{q} \in \mathcal{Q}$, denoted by $\widetilde{\nabla}\mathcal{I}(\mathbf{q})$; i.e., $\widetilde{\nabla}\mathcal{I}(\mathbf{q}) \in \partial\mathcal{I}(\mathbf{q})$. Thus, the subdifferential inclusion condition in (10b) can be replaced by

$$\nabla g\big(\mathbf{q}^\star\big) - \boldsymbol{\lambda}^\star + \widetilde{\nabla}\mathcal{I}\big(\mathbf{q}^\star\big) = \mathbf{0}, \text{ for some } \widetilde{\nabla}\mathcal{I}\big(\mathbf{q}^\star\big). \tag{11}$$

The ensuing analysis will rely on (11) as a more tractable version of (10b). This subgradient based treatment of KKT conditions is popular among many recent references including [32], [33]. The subgradient $\widetilde{\nabla}\mathcal{I}(\mathbf{q}^\star)$ in (11) can be expressed as $\boldsymbol{\lambda}^\star - \nabla g(\tilde{\mathbf{q}}^\star)$ where

$$\tilde{\mathbf{q}}^\star \triangleq \arg\min_{\mathbf{q}\in\mathcal{Q}} g(\mathbf{q}) - \langle \boldsymbol{\lambda}^\star, \mathbf{q}\rangle.$$

Note that $\tilde{\mathbf{q}}^\star$ is well-defined when $g(\mathbf{q})$ is strongly convex or $\mathcal{Q}$ is compact, exactly the case in our problem. Using these notations, the PPD updates in (S1)-(S3) are equivalent to

$$\begin{aligned}
&\mathbf{v}\text{-update: } \mathbf{v}^{k+1} = \arg\min_{\mathbf{v}} f(\mathbf{v}) + \langle \boldsymbol{\lambda}^k, \mathbf{B}\mathbf{v}\rangle;\\
&\mathbf{q}\text{-update: } \mathbf{q}^{k+1} = \Big\{\mathbf{q}^k - \alpha\nabla g\Big(\mathbf{q}^k\Big) + \alpha\boldsymbol{\lambda}^k\Big\};\\
&\boldsymbol{\lambda}\text{-update: } \boldsymbol{\lambda}^{k+1} = \boldsymbol{\lambda}^k + \beta\Big(\mathbf{B}\mathbf{v}^{k+1} - \mathbf{q}^{k+1} - \mathbf{w}\Big).
\end{aligned} \tag{12}$$

The recursive relation of the iterate $\{\mathbf{v}^k, \mathbf{q}^k, \boldsymbol{\lambda}^k\}$ in (12) can be further written as[1]

$$\nabla f\Big(\mathbf{v}^{k+1}\Big) + \mathbf{B}^\top\boldsymbol{\lambda}^k = \mathbf{0}, \tag{13a}$$

$$\mathbf{q}^{k+1} = \mathbf{q}^k - \alpha\nabla g\Big(\mathbf{q}^k\Big) + \alpha\boldsymbol{\lambda}^k - \alpha\widetilde{\nabla}\mathcal{I}\Big(\mathbf{q}^{k+1}\Big), \tag{13b}$$

$$\boldsymbol{\lambda}^{k+1} = \boldsymbol{\lambda}^k + \beta\Big(\mathbf{B}\mathbf{v}^{k+1} - \mathbf{q}^{k+1} - \mathbf{w}\Big). \tag{13c}$$

The gist of our analysis is to show all first-order residuals of (13), namely $\|\nabla f(\mathbf{v}^{k+1}) + \mathbf{B}^\top\boldsymbol{\lambda}^k\|$, $\|\mathbf{q}^k - \mathbf{q}^{k+1}\|_2^2$, and $\|\boldsymbol{\lambda}^k - \boldsymbol{\lambda}^{k+1}\|_2^2$, will asymptotically converge to zero. Due to the existence and uniqueness of $(\mathbf{v}^\star, \mathbf{q}^\star)$ under strong convexity, a vanishing property of first-order residuals leads to the asymptotic convergence of the iterates $(\mathbf{v}^k, \mathbf{q}^k)$. The conditions of step-size choice will be given here for achieve the vanishing property.

Theorem 1: Let $\widetilde{\eta}$ and $\widetilde{L}$ be the smallest and largest singular values of matrix $\mathbf{B}$, respectively. If the positive α and β are chosen such that

$$\begin{cases} \alpha < 2/\big[\gamma\big(\widetilde{L}^{-1} + \widetilde{\eta}^{-1}\big)\big],\\ \beta < 2/\big[\widetilde{L}^2 + \gamma^{-1}\big(\widetilde{L} + \widetilde{\eta}\big)\big],\end{cases} \tag{14}$$

then the sequence $\{\mathbf{q}^k\}$ generated by the PPD updates (S1)-(S3) converges to the optimum $\mathbf{q}^\star$.

[1] Using the subgradient and indicator function, one can write a general gradient projection update as a subgradient update.

Proof: We first show that the successive difference $\{\|\mathbf{q}^k - \mathbf{q}^{k+1}\|_2^2 + \|\boldsymbol{\lambda}^k - \boldsymbol{\lambda}^{k+1}\|_2^2\}$ is an infinitely summable sequence and thus converges to zero. We will form the difference between the iterates and their corresponding optimum solutions. To this end, substituting the update (13c) for $\boldsymbol{\lambda}^k$ into (13a) and (13b), respectively, and then subtracting the two latter equations by the KKT conditions (10)-(11), we have

$$\begin{aligned}
&\nabla f\Big(\mathbf{v}^{k+1}\Big) - \nabla f\big(\mathbf{v}^\star\big) + \mathbf{B}^\top\Big(\boldsymbol{\lambda}^{k+1} - \boldsymbol{\lambda}^\star\Big)\\
&\quad - \beta\mathbf{B}^\top\Big(\mathbf{B}\Big(\mathbf{v}^{k+1} - \mathbf{v}^\star\Big) - \Big(\mathbf{q}^{k+1} - \mathbf{q}^\star\Big)\Big) = \mathbf{0},
\end{aligned} \tag{15a}$$

$$\begin{aligned}
\mathbf{q}^{k+1} &= \mathbf{q}^k - \alpha\Big(\nabla g\Big(\mathbf{q}^k\Big) - \nabla g\big(\mathbf{q}^\star\big)\Big) + \alpha\Big(\boldsymbol{\lambda}^{k+1} - \boldsymbol{\lambda}^\star\Big)\\
&\quad - \alpha\beta\Big(\mathbf{B}\Big(\mathbf{v}^{k+1} - \mathbf{v}^\star\Big) - \Big(\mathbf{q}^{k+1} - \mathbf{q}^\star\Big)\Big)\\
&\quad - \alpha\Big(\widetilde{\nabla}\mathcal{I}\Big(\mathbf{q}^{k+1}\Big) - \widetilde{\nabla}\mathcal{I}\big(\mathbf{q}^\star\big)\Big),
\end{aligned} \tag{15b}$$

$$\boldsymbol{\lambda}^{k+1} = \boldsymbol{\lambda}^k + \beta\Big(\mathbf{B}\Big(\mathbf{v}^{k+1} - \mathbf{v}^\star\Big) - \Big(\mathbf{q}^{k+1} - \mathbf{q}^\star\Big)\Big). \tag{15c}$$

Additionally, by the strong convexity and gradient Lipschitz continuity of $g(\cdot)$, we have

$$\begin{aligned}
&\frac{2\eta L}{\eta + L}\Big\|\mathbf{q}^k - \mathbf{q}^\star\Big\|_2^2 + \frac{2}{\eta + L}\Big\|\nabla g\Big(\mathbf{q}^k\Big) - \nabla g\big(\mathbf{q}^\star\big)\Big\|_2^2\\
&\quad \le 2\Big\langle \mathbf{q}^k - \mathbf{q}^\star, \nabla g\Big(\mathbf{q}^k\Big) - \nabla g\big(\mathbf{q}^\star\big)\Big\rangle\\
&\quad \le 2\Big\langle \mathbf{q}^{k+1} - \mathbf{q}^\star, \nabla g\Big(\mathbf{q}^k\Big) - \nabla g\big(\mathbf{q}^\star\big)\Big\rangle\\
&\qquad + \frac{\eta + L}{2}\Big\|\mathbf{q}^k - \mathbf{q}^{k+1}\Big\|_2^2 + \frac{2}{\eta + L}\Big\|\nabla g\Big(\mathbf{q}^k\Big) - \nabla g\big(\mathbf{q}^\star\big)\Big\|_2^2.
\end{aligned} \tag{16}$$

The first inequality of (16) is a standard result under strong convexity and gradient Lipschitz continuity; see [34, Th. 2.1.11]. Reorganizing (16) gives rise to

$$\begin{aligned}
\frac{2\alpha\eta L}{\eta + L}\Big\|\mathbf{q}^k - \mathbf{q}^\star\Big\|_2^2 &\le 2\Big\langle \mathbf{q}^{k+1} - \mathbf{q}^\star, \alpha\Big(\nabla g\Big(\mathbf{q}^k\Big) - \nabla g\big(\mathbf{q}^\star\big)\Big)\Big\rangle\\
&\quad + \frac{\alpha(\eta + L)}{2}\Big\|\mathbf{q}^k - \mathbf{q}^{k+1}\Big\|_2^2.
\end{aligned} \tag{17}$$

Substituting (15b) into (17) for $\alpha(\nabla g(\mathbf{q}^k) - \nabla g(\mathbf{q}^\star))$ leads to

$$\begin{aligned}
&\frac{2\alpha\eta L}{\eta + L}\Big\|\mathbf{q}^k - \mathbf{q}^\star\Big\|_2^2\\
&\le 2\Big\langle \mathbf{q}^{k+1} - \mathbf{q}^\star, \mathbf{q}^k - \mathbf{q}^{k+1}\Big\rangle + 2\Big\langle \mathbf{q}^{k+1} - \mathbf{q}^\star, \alpha\boldsymbol{\lambda}^{k+1} - \alpha\boldsymbol{\lambda}^\star\Big\rangle\\
&\quad - 2\Big\langle \mathbf{q}^{k+1} - \mathbf{q}^\star, \alpha\beta\mathbf{B}\Big(\mathbf{v}^{k+1} - \mathbf{v}^\star\Big)\Big\rangle\\
&\quad + 2\Big\langle \mathbf{q}^{k+1} - \mathbf{q}^\star, \alpha\beta\Big(\mathbf{q}^{k+1} - \mathbf{q}^\star\Big)\Big\rangle\\
&\quad - 2\Big\langle \mathbf{q}^{k+1} - \mathbf{q}^\star, \alpha\Big(\widetilde{\nabla}\mathcal{I}\Big(\mathbf{q}^{k+1}\Big) - \widetilde{\nabla}\mathcal{I}\big(\mathbf{q}^\star\big)\Big)\Big\rangle\\
&\quad + \frac{\alpha(\eta + L)}{2}\Big\|\mathbf{q}^k - \mathbf{q}^{k+1}\Big\|_2^2\\
&\le 2\Big\langle \mathbf{q}^{k+1} - \mathbf{q}^\star, \mathbf{q}^k - \mathbf{q}^{k+1}\Big\rangle + 2\alpha\Big\langle \mathbf{q}^{k+1} - \mathbf{q}^\star, \boldsymbol{\lambda}^{k+1} - \boldsymbol{\lambda}^\star\Big\rangle\\
&\quad - 2\alpha\beta\Big\langle \mathbf{q}^{k+1} - \mathbf{q}^\star, \mathbf{B}\Big(\mathbf{v}^{k+1} - \mathbf{v}^\star\Big)\Big\rangle + 2\alpha\beta\Big\|\mathbf{q}^{k+1} - \mathbf{q}^\star\Big\|_2^2\\
&\quad + \frac{\alpha(\eta + L)}{2}\Big\|\mathbf{q}^k - \mathbf{q}^{k+1}\Big\|_2^2.
\end{aligned} \tag{18}$$

As for the function $f(\cdot)$, by the strong convexity we have

$$2c\Big\|\mathbf{v}^{k+1} - \mathbf{v}^\star\Big\|_2^2 \le 2\Big\langle \mathbf{v}^{k+1} - \mathbf{v}^\star, \nabla f\Big(\mathbf{v}^{k+1}\Big) - \nabla f\big(\mathbf{v}^\star\big)\Big\rangle. \tag{19}$$

Substituting $\nabla f(\mathbf{v}^{k+1}) - \nabla f(\mathbf{v}^\star)$ of (15a) into (19) leads to

$$\begin{aligned} &2\alpha c\left\|\mathbf{v}^{k+1}-\mathbf{v}^\star\right\|_2^2 \\ &\le 2\alpha\Big\langle \mathbf{v}^{k+1}-\mathbf{v}^\star, \mathbf{B}^\top\Big(\boldsymbol{\lambda}^\star-\boldsymbol{\lambda}^{k+1}\Big) \\ &\quad + \beta\mathbf{B}^\top\Big(\mathbf{B}\Big(\mathbf{v}^{k+1}-\mathbf{v}^\star\Big)-\Big(\mathbf{q}^{k+1}-\mathbf{q}^\star\Big)\Big)\Big\rangle \\ &= 2\alpha\Big\langle\mathbf{B}\Big(\mathbf{v}^{k+1}-\mathbf{v}^\star\Big), \boldsymbol{\lambda}^\star-\boldsymbol{\lambda}^{k+1}\Big\rangle + 2\alpha\beta\left\|\mathbf{v}^{k+1}-\mathbf{v}^\star\right\|_{\mathbf{B}^\top\mathbf{B}}^2 \\ &\quad - 2\alpha\beta\Big\langle\mathbf{B}\Big(\mathbf{v}^{k+1}-\mathbf{v}^\star\Big), \mathbf{q}^{k+1}-\mathbf{q}^\star\Big\rangle. \end{aligned} \tag{20}$$

Summing up (18) and (20) results in

$$\begin{aligned} &\frac{2\alpha\eta L}{\eta+L}\left\|\mathbf{q}^k-\mathbf{q}^\star\right\|_2^2 + 2\alpha c\left\|\mathbf{v}^{k+1}-\mathbf{v}^\star\right\|_2^2 \\ &\le 2\Big\langle\mathbf{q}^{k+1}-\mathbf{q}^\star, \mathbf{q}^k-\mathbf{q}^{k+1}\Big\rangle + 2\alpha\Big\langle\mathbf{q}^{k+1}-\mathbf{q}^\star, \boldsymbol{\lambda}^{k+1}-\boldsymbol{\lambda}^\star\Big\rangle \\ &\quad - 2\alpha\beta\Big\langle\mathbf{q}^{k+1}-\mathbf{q}^\star, \mathbf{B}\Big(\mathbf{v}^{k+1}-\mathbf{v}^\star\Big)\Big\rangle + 2\alpha\beta\left\|\mathbf{q}^{k+1}-\mathbf{q}^\star\right\|_2^2 \\ &\quad + \frac{\alpha(\eta+L)}{2}\left\|\mathbf{q}^k-\mathbf{q}^{k+1}\right\|_2^2 \\ &\quad + 2\alpha\Big\langle\mathbf{B}\Big(\mathbf{v}^{k+1}-\mathbf{v}^\star\Big), \boldsymbol{\lambda}^\star-\boldsymbol{\lambda}^{k+1}\Big\rangle + 2\alpha\beta\left\|\mathbf{v}^{k+1}-\mathbf{v}^\star\right\|_{\mathbf{B}^\top\mathbf{B}}^2 \\ &\quad - 2\alpha\beta\Big\langle\mathbf{B}\Big(\mathbf{v}^{k+1}-\mathbf{v}^\star\Big), \mathbf{q}^{k+1}-\mathbf{q}^\star\Big\rangle. \end{aligned} \tag{21}$$

Consider a basic inequality rule for Euclidean norm

$$2\Big\langle\sqrt{\rho}\mathbf{a}, \sqrt{\rho^{-1}}\mathbf{b}\Big\rangle \le \rho\|\mathbf{a}\|^2 + \rho^{-1}\|\mathbf{b}\|^2,$$

which holds for any $\rho > 0$ and vectors $\mathbf{a}$ and $\mathbf{b}$ of the same dimension. Using this inequality and (15c), the right-hand-side of (21) can be relaxed by

$$\begin{aligned} &2\Big\langle\mathbf{q}^{k+1}-\mathbf{q}^\star, \mathbf{q}^k-\mathbf{q}^{k+1}\Big\rangle + \frac{2\alpha}{\beta}\Big\langle\boldsymbol{\lambda}^k-\boldsymbol{\lambda}^{k+1}, \boldsymbol{\lambda}^{k+1}-\boldsymbol{\lambda}^\star\Big\rangle \\ &\quad + \left(\frac{(1-\varepsilon)\alpha}{\beta}+\frac{(1+\varepsilon)\alpha}{\beta}\right)\left\|\boldsymbol{\lambda}^k-\boldsymbol{\lambda}^{k+1}\right\|_2^2 \\ &\quad + \frac{\alpha(\eta+L)}{2}\left\|\mathbf{q}^k-\mathbf{q}^{k+1}\right\|_2^2 \\ &\le \left\|\mathbf{q}^k-\mathbf{q}^\star\right\|_2^2 - \left\|\mathbf{q}^{k+1}-\mathbf{q}^\star\right\|_2^2 - \left\|\mathbf{q}^k-\mathbf{q}^{k+1}\right\|_2^2 \\ &\quad + \frac{\alpha}{\beta}\left(\left\|\boldsymbol{\lambda}^k-\boldsymbol{\lambda}^\star\right\|_2^2 - \left\|\boldsymbol{\lambda}^{k+1}-\boldsymbol{\lambda}^\star\right\|_2^2\right) - \varepsilon\left\|\boldsymbol{\lambda}^k-\boldsymbol{\lambda}^{k+1}\right\|_2^2 \\ &\quad + \frac{\alpha(\eta+L)}{2}\left\|\mathbf{q}^k-\mathbf{q}^{k+1}\right\|_2^2 \\ &\quad + (1+\varepsilon)\alpha\beta(1+\rho)\left\|\mathbf{B}\Big(\mathbf{v}^{k+1}-\mathbf{v}^\star\Big)\right\|_2^2 \\ &\quad + (1+\varepsilon)\alpha\beta\left(1+\frac{1}{\rho}\right)\left\|\mathbf{q}^{k+1}-\mathbf{q}^\star\right\|_2^2 \end{aligned} \tag{22}$$

where ε is any arbitrary constant within (0, 1). It then follows from (21) and (22) that

$$\begin{aligned} &\left(1-\frac{\alpha(\eta+L)}{2}\right)\left\|\mathbf{q}^k-\mathbf{q}^{k+1}\right\|_2^2 + \varepsilon\left\|\boldsymbol{\lambda}^k-\boldsymbol{\lambda}^{k+1}\right\|_2^2 \\ &\le \left(1-\frac{2\alpha\eta L}{\eta+L}\right)\left\|\mathbf{q}^k-\mathbf{q}^\star\right\|_2^2 + \frac{\alpha}{\beta}\left(\left\|\boldsymbol{\lambda}^k-\boldsymbol{\lambda}^\star\right\|_2^2 - \left\|\boldsymbol{\lambda}^{k+1}-\boldsymbol{\lambda}^\star\right\|_2^2\right) \\ &\quad - \left(1-(1+\varepsilon)\alpha\beta\left(1+\frac{1}{\rho}\right)\right)\left\|\mathbf{q}^{k+1}-\mathbf{q}^\star\right\|_2^2 \\ &\quad - \Big(2\alpha c-(1+\varepsilon)\alpha\beta(1+\rho)\sigma_{\max}\left\|\mathbf{v}^{k+1}-\mathbf{v}^\star\right\|_2^2 \end{aligned}$$

where $\sigma_{\max}$ is defined as the largest singular value of $\mathbf{B}^\top\mathbf{B}$. This inequality is crucial for showing the iterative difference is infinitely summable. By comparing the coefficients corresponding to each term, we come up with the following set of necessary conditions:

$$\begin{cases} 1-\frac{\alpha(\eta+L)}{2} > 0, \\ \varepsilon > 0, \\ 1-\frac{2\alpha\eta L}{\eta+L} \le 1-(1+\varepsilon)\alpha\beta\Big(1+\frac{1}{\rho}\Big), \\ 2\alpha c-(1+\varepsilon)\alpha\beta(1+\rho)\sigma_{\max} \ge 0, \end{cases} \tag{23}$$

which is equivalent to having

$$\begin{cases} \alpha < \frac{2}{\eta+L}, \\ \beta < \min\left\{\frac{2\eta L}{(\eta+L)\left(1+\frac{1}{\rho}\right)}, \frac{2c}{(1+\rho)\sigma_{\max}}\right\}. \end{cases} \tag{24}$$

The best range of β is achieved by setting $\rho = \frac{c(\eta+L)}{\eta L\sigma_{\max}}$. Accordingly, the step-size rule for β becomes

$$\beta < \frac{2c\eta L}{\eta L\sigma_{\max}+c(\eta+L)}. \tag{25}$$

The choice of α and β given by (24)-(25) guarantees the infinite summability of the sequence $\{\|\mathbf{q}^k-\mathbf{q}^{k+1}\|_2^2+\|\boldsymbol{\lambda}^k-\boldsymbol{\lambda}^{k+1}\|_2^2\}$. Hence, each one of the summands, namely $\|\mathbf{q}^k-\mathbf{q}^{k+1}\|$ and $\|\boldsymbol{\lambda}^k-\boldsymbol{\lambda}^{k+1}\|_2$, converges to zero. Based on (13), we thus assert

$$\left\|\nabla f\Big(\mathbf{v}^{k+1}\Big)+\mathbf{B}^\top\boldsymbol{\lambda}^k\right\|_2 = 0, \tag{26a}$$

$$\lim_{k\to\infty}\left\|\nabla g\Big(\mathbf{q}^k\Big)-\boldsymbol{\lambda}^k+\widetilde{\nabla}\mathcal{I}\Big(\mathbf{q}^{k+1}\Big)\right\|_2 = 0, \tag{26b}$$

$$\lim_{k\to\infty}\left\|\mathbf{B}\mathbf{v}^{k+1}-\mathbf{q}^{k+1}-\mathbf{w}\right\|_2 = 0. \tag{26c}$$

Comparing (26) with (10)-(11), one can conclude that the KKT conditions hold for the limit and thus the sequence $\{\mathbf{q}^k\}$ converges to $\mathbf{q}^\star$.

Last, for the specific forms of $f = f_1$ and $g = f_2$, their respective Hessian is $\nabla^2 f = \mathbf{I}$ and $\nabla^2 g = \gamma\mathbf{X}$. Since $\widetilde{\eta}$ and $\widetilde{L}$ denote the smallest and largest singular values of $\mathbf{B}$, respectively, we have $\eta = \gamma\widetilde{L}^{-1}$ and $L = \gamma\widetilde{\eta}^{-1}$ for its inverse $\mathbf{X}$. As $c = 1$ for the function f_1, the step-size conditions in (25) are equivalent to (14). ■

The step-size rule (14) suggests $\alpha = O(\widetilde{\eta})$ and $\beta = O(1/\widetilde{L}^2)$. In general, a larger network size increases the value of $\widetilde{L}$, and thus may reduce the range of β. Fortunately, our empirical experience suggests that the choice of β minimally affects the convergence speed, while the step-size α for $\mathbf{q}$-update plays a more important role. As for the smallest eigenvalue $\widetilde{\eta}$ for the reduced graph Laplacian, it is strongly related to the connectivity[2] of the network. As long as the network is connected, this quantity is lower bounded away from zero. Empirically, a larger α value leads to increased convergence speed. Thus, a more clustered power network would result in a faster convergence, which coincides with the common wisdom. A thorough investigation of convergence

[2]Note that the algebraic connectivity of the graph is defined as the second smallest eigenvalue of the standard Laplacian. Thus, the exact relation between the algebraic connectivity and $\widetilde{\eta}$ needs further investigation in future.

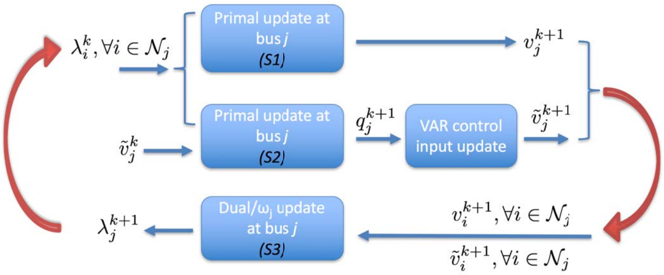

Fig. 1. Online implementation of the proposed HVC design that adapts to dynamic system conditions and constantly updates the VAR control inputs. Rectangular blocks denote local computational tasks of PPD iterations and control updates, while the two vertical arrows correspond to the communications exchange among bus j and its neighboring buses.

rate would involve the spectral analysis on the eigenvalues of matrix $\mathbf{B}$, which is an interesting future research direction.

Remark 2 (General Error Objective Functions): Our convergence analysis is performed by assuming both functions f and g are strongly convex. This assumption can be relaxed to the so-termed restricted strongly convex functions [35], such as the Huber loss function. To this end, the first inequality in (16) needs to be modified to have more conservative coefficients on the left-hand-side, leading to a narrower region of step-size choice. In addition, it is possible to completely remove the (restricted) strong convexity assumption on f_1. Specifically, the $\mathbf{v}$-update in (12) can be simply modified as

$$\mathbf{v}^{k+1} = \arg\min_{\mathbf{v}} f_1(\mathbf{v}) + \left\langle \boldsymbol{\lambda}^k, \mathbf{B}\mathbf{v} \right\rangle + \vartheta \left\| \mathbf{v} - \mathbf{v}^k \right\|^2$$

to tackle the case of lacking (restricted) strong convexity. Compared to the previous update scheme, one more proximal term is augmented to stabilize the system. Informally speaking, the additional term $\vartheta \|\mathbf{v} - \mathbf{v}^k\|^2$ can enhance the stability region of step-sizes for non-strongly convex functions. In this case, the step-size rule would depend on the choice of ϑ. Although the modified update would allow more general error objective functions, such generalization is likely to reduce convergence speed in practice and thus is not employed in our current HVC strategy.

B. Online Feedback Design

To extend to dynamic operating conditions, we develop an algorithm for bus j to update w_j using neighborhood voltage information. As mentioned earlier, each w_j depends on the network-wide active and reactive power injections, and thus it is complicated to directly compute it locally at bus j. Interestingly, we can leverage the sparsity structure of $\mathbf{B}$ in (1) to estimate the w_j^{k+1} at iteration $(k+1)$ from neighboring voltage information only. Upon measuring its voltage $\widetilde{v}_j^{k+1}$ in (S2) and receiving the neighboring $\{\widetilde{v}_i^{k+1}\}_{i \in \mathcal{N}_j}$, bus j can update

$$w_j^{k+1} := \sum_{i \in \mathcal{N}_j} B_{ji} \widetilde{v}_i^{k+1} - q_j^{k+1}, \tag{27}$$

which is used for the $\boldsymbol{\lambda}$-update in (S3). This feedback-based update (27) is again based on the linearized power flow $\mathbf{B}\widetilde{\mathbf{v}}^{k+1} \approx \mathbf{q}^{k+1} + \mathbf{w}^{k+1}$. Note that each bus only needs the knowledge of the reactance for all incident lines in (27). The advantages of this feedback design are three-fold. First, each bus j subsumes exactly the same bus-to-bus communication network as in the distributed PPD updates. Second, the voltage measurements contain the latest system information and effectively track the dynamically varying operating conditions. Last, a feedback control design in general improves the robustness to system modeling mismatch (see [36, Sec. 8.9]). Hence, the actual voltage measurements used i (27) can potentially capture the underlying non-linearity of power flow model, as demonstrated by our numerical results in Section IV-A. Fig. 1 offers a schematic of the proposed PPD-based HVC implementation. Every iteration consists of the *local computations* of (S1)-(S3) per bus j (the blocks), in addition to two steps of information exchange among all neighboring buses (the vertical arrows). All the computational tasks for (S1)-(S3) are simple linear updates and thus can be executed efficiently. After updating q_j^{k+1} in (S2), bus j inputs it as the control signal for the local VAR resource and then measures and broadcasts the voltage $\tilde{v}_j^{k+1}$ to neighboring buses for updating ω_j^{k+1} in (27).

Remark 3 (Cyber Network Topology): The bus-to-bus architecture of the HVC design can be generalized to coordinate clusters of buses as long as the cyber network is still connected; see [9]. This way, it is not necessary to have the DERs to be connected to each other by a line segment. For the buses without any controllable DERs connected, we may also eliminate them to create an equivalent network using the Kron Reduction method [37, Sec. 9.3]. The resultant equivalent network would consist of those DER-connected buses only. The voltage-based feedback update in (27) can be generalized as well using the reduced *Bbus matrix* for the equivalent network. To sum up, the proposed HVC design allows for flexible cyber network topology and can be easily generalized to distribution networks where DERs are not connected to each and every bus.

C. Limited Communication Rates

The performance of the proposed HVC design depends on the quality of communication links, which has been assumed to be perfect so far. However, random link failures and messaging delays are common in contemporary digital communication systems because of either network congestion or poor signal-to-noise ratios in some wireless environments. It is imperative to examine how the PPD-based HVC scheme would perform under imperfect communication. By using the handshaking protocol and embedding time-stamp information into messages to be sent to adjacent buses, buses can reliably exchange information, recognize delays, and treat significantly delayed packets as of link (retrieval) failure. Such effect caused by communicational imperfectness is summarized as the so-termed "asynchrony" in decentralized computational networks [38]. A number of references have appeared in the literature of distributed computing to deal with such issue by allowing delays and/or time-varying network topology in the numerical (optimization) algorithms [39]–[46]. Informally speaking, as long as the time-varying communication network

Algorithm 1 The A-HVC Update Steps (AS1)-(AS3)

for every iteration $k = 1, 2, \dots$ **do**
  for bus $j \in \mathcal{N}_a^k$ **do**
    (AS1): update v_j^{k+1} as in (4);
    (AS2): update q_j^{k+1} as in (7), and w_j^{k+1} as in (27);
    (AS3): update λ_j^{k+1} as in (8).
  end for
  for bus $j \notin \mathcal{N}_a^k$ **do**
    (AS1): set $v_j^{k+1} = v_j^k$;
    (AS2): update q_j^{k+1} as in (7), and set $w_j^{k+1} = w_j^k$;
    (AS3): set $\lambda_j^{k+1} = \lambda_j^k$.
  end for
end for

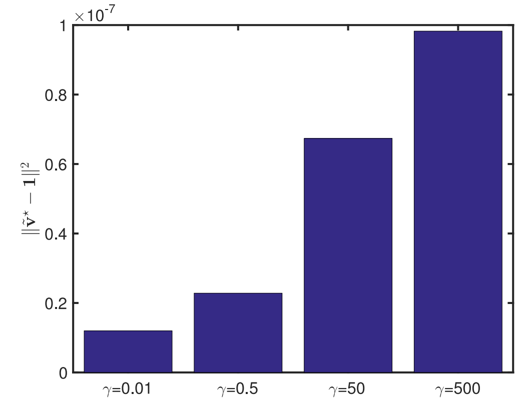

Fig. 2. Voltage mismatch squared error versus various values of parameter γ under the static system setting.

is connected on average[3] and the delay is finite, one could choose more conservative step-size values for stability concerns. A rigorous proof of convergence properties under these conditions is out of the scope of this paper and will be a future direction.

To tackle the link failure scenario, we adopt the "freezing" strategy from earlier work [39]–[41]. Conceptually, each primal or dual variable would remain unchanged until new information becomes available for its update from neighboring buses. For a related primal-dual algorithm, [41] has shown the convergence of its asynchronous version under random activation of agents, under which a link $(i, j) \in \mathcal{E}$ is activated only if both nodes i and j are. The activation of each node has been assumed to follow a Bernoulli distribution, independent across time and nodes. Nonetheless, the specific asynchronous algorithm developed in [41] would freeze all variables and thus fail to work for a total communication outage. Thus, we have developed a different asynchronous (A-)HVC algorithm to ensure sufficient adaptability even during total link failures. For each inactive bus j, the A-HVC algorithm works by freezing its associated optimization-related variables v_j and λ_j while constantly updating the VAR variable q_j using (7). Even if there is no information at all from the neighboring buses, the gradient direction of f_2 for bus j can be always obtained from $\tilde{v}_j^k$. This is exactly the reason for including the weighted error objective f_2. Under a total communication outage, the A-HVC algorithm boils down to a surrogate local control design with the set-point depending on the latest updated λ_j^k. This special case is very related to the secondary frequency/voltage control in microgrids where the dual variables provide the set-points of the local droop curves; see [22], [47]. By denoting $\mathcal{N}_a^k \subseteq \mathcal{N}$ as the subset of activated nodes per iteration k, the proposed A-HVC updates are tabulated in Algorithm 1. Note that due to the first order nature of (AS2), the transient level of our A-HVC algorithm depends mostly on the intensity of external disturbances and very minimally on the availability of communication signals.

[3]The network is not necessarily connected at every step. Please refer to [45, Assumption 2] for the definition of "B-connectivity" of time-varying graphs.

IV. Numerical Tests

We perform several numerical tests to demonstrate the effectiveness of the proposed communication-cognizant HVC design for a simple feeder system and a realistic distribution system. Numerical performance results are provided for both static and dynamic operating conditions. The dynamic tests are performed on the IEEE 123-bus test case [48]. The desired voltage magnitude μ_j is chosen to be 1 p.u. at every bus j. Albeit the HVC design is developed using the linearized model (1), all presented test results have been obtained by solving the full ac (unbalanced) power flow equations. All numerical tests are implemented in the MathWorks MATLAB 2014a simulator, and the OpenDSS package is used for solving the multi-phase unbalanced ac power flow. Therefore, the actual bus voltage $\tilde{\mathbf{v}}$, instead of the output of (1), is used to update the VAR variables as in (7) and to quantify the error performance.

A. Static System Operating Conditions

A single-phase radial power distribution feeder that consists of 21 buses is first used for static tests. This network has $N = 20$ and the impedance of each line segment is set to be $(0.233 + j0.366)\Omega$. As mentioned earlier, the voltage at the feeder head is assumed perfectly regulated by the OLTC and thus fixed at 1 p.u.; i.e., $v_0 = 1$. Per bus j, we set the loading with $p_j^c = 70$kW and $q_j^c = 20$kVAR. Also, the inverter rating is set to be $(70 + \psi_j)$kVA with ψ_j being a random realization of a Gaussian distributed variable of zero mean and standard deviation at 13.33, thus modeling a roughly 50% variation in inverter sizing according to the three-sigma rule of thumb. Under these settings, the VAR constraints (2c) are actively enforced for some locations. We test the HVC algorithm using various γ values to illustrate the potential trade-off between distributed and local updates. Fig. 2 plots the voltage mismatch squared error norm $\|\tilde{\mathbf{v}}^\star - \mathbf{1}\|^2$ obtained for each γ value while keeping all other settings exactly the same. As expected, the larger γ is, the more pronounced the voltage deviation is. This coincides with our claim that under limited VAR resources the weighted error objective f_2 can affect the optimality of the resultant HVC solutions. For given system settings and rates

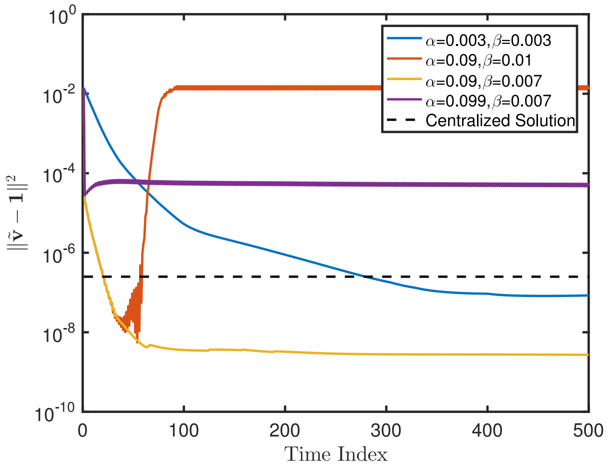

Fig. 3. Log-scale voltage mismatch squared error versus the number of iterations for four cases of step-size choice under the static system setting.

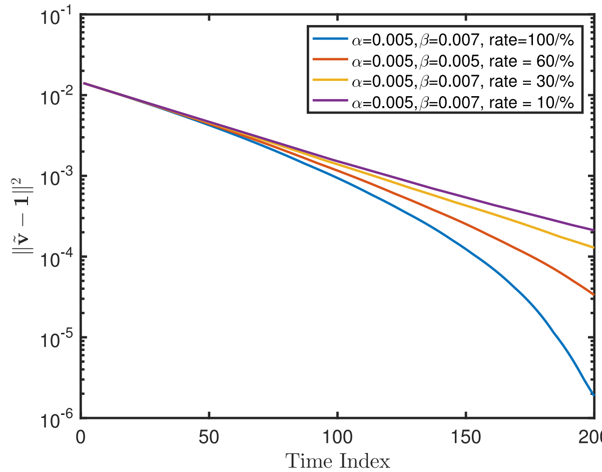

Fig. 4. Log-scale voltage mismatch squared error versus the number of iterations for various bus activation rate under fixed step-size choices and the static system setting.

of communications, it is possible perform offline studies to better select the γ value. We pick the value $\gamma = 0.5$ for the ensuing tests.

Based on Theorem 1, we can compute the step-size bounds to be $\alpha < 0.092$ and $\beta < 0.0073$. To verify the bounds, we test the HVC algorithm with four different cases of steps-size choices and plot the corresponding log-scale iterative squared error $\|\tilde{\mathbf{v}}-\mathbf{1}\|^2$ in Fig. 3, all under perfect communication links. Hence, the case of $\alpha = 0.09$ and $\beta = 0.007$ is guaranteed convergent under Theorem 1, along with a more conservative case of $\alpha = 0.003$ and $\beta = 0.003$. Either $\alpha = 0.099$ or $\beta = 0.01$ would violate the analytical step-size bounds. For the last two cases, the PPD updates are shown to fail to converge to the optimal VAR solutions. Note that the iterative error grows up until the VAR limits are active everywhere, resulting in a constant floor above the optimal value. This has verified the value of Theorem 1 for selecting the step-size. Between the first two convergent cases, it has been observed that a larger value of step-size would be preferred for accelerating the convergence rate. Nonetheless, a more aggressive step-size choice would more easily lead to oscillations or even instability concerns when operating conditions or system configurations vary. To tackle this problem, it is possible to design an adaptive strategy to select the step-size by observing the oscillation level of voltage. Last, Fig. 3 also includes the benchmark performance by solving the centralized problem (2) with $\mathbf{w}$ given. Interestingly, our HVC approach, once convergent, has achieved a even smaller steady-state mismatch error than the centralized solution. This is exactly because of the feedback design that has used the actual voltage measurements to mitigate the modeling mismatch from the linearized power flow approximation. To sum up, Fig. 3 has validated our analytical step-size bounds and demonstrated the strength of our feedback-based control design.

To demonstrate the robustness against link failures, we have tested our A-HVC algorithm for different bus activation rates. For a given rate, every bus activates randomly at each iteration following a Bernoulli distribution, independent from each other. For this test, the step-sizes are set to be $\alpha = 0.005$ and $\beta = 0.007$, guaranteeing the convergence. Fig. 4 plots the log-scale iterative squared error with the activation rate ranging from 10% to 100%, with a 100% rate corresponding to the perfectly synchronized communication scenario. Asymptotic convergence has been observed for any activation rate, while the convergence speed decreases with more frequent link failures. Intuitively, when the activation rate increases the voltage information can be diffused more effectively across the network, with the fully synchronous case exhibiting the fastest convergence.

B. Dynamic System Operating Conditions

To corroborate our HVC scheme for online implementation, we have tested the proposed algorithm on the IEEE 123-bus test case [48]. Dynamic system operating conditions are generated using real load profiles from an online data repository [49], as shown in Fig. 5. The minute-sampled active and reactive power consumption data along with solar profile were collected on Friday, June 20, 2010 for residential loads. Each home shares similar load and generation patterns, which are diversified by small random additive noises. For each load node of the 123-bus test case, it is connected to a certain number of residential homes where each home is equipped with solar generations of 3.5kVA rated capacity. The number of homes at each node is determined by rounding up the ratio between the active spot load demand in the original case and the maximum daily active load demand (6kW) of a residential home (as depicted by Fig. 5). Accordingly, DERs are connected to every node with active loading in our revised 123-bus case. In addition, at every minute period, the VAR limits $[\underline{\mathbf{q}}, \overline{\mathbf{q}}]$ are updated based on the instantaneous active power from solar generations using the inverter rating (i.e., 3.5kVA per inverter). Under these settings, the inverter VAR resources turn out to be insufficient to achieve perfectly flat voltage at all times. This implies the VAR constraints are actively enforced in (2) during the operations.

Fig. 6 plots the daily network-wide voltage mismatch squared error norm for the a-phase of the 123-bus. The plots

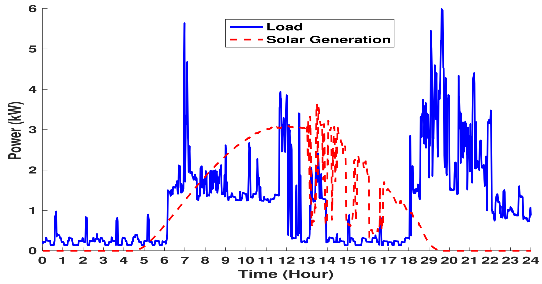

Fig. 5. Sample daily load and solar generation profiles of a single residential home.

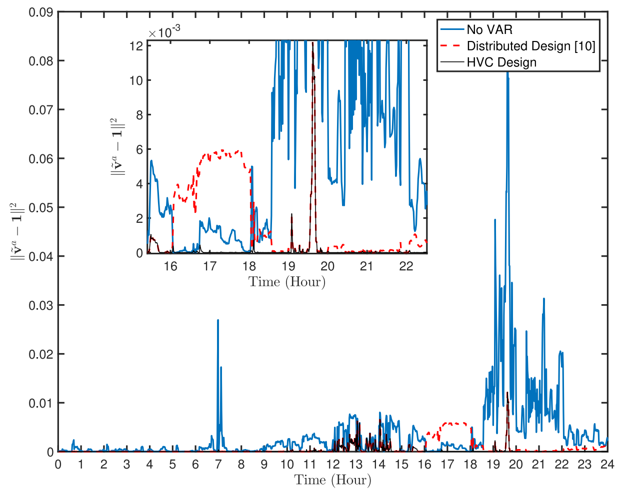

Fig. 6. Daily voltage mismatch squared error for a-phase of the 123-bus under three different control strategies. A total communication link failure occurs from hour 16:00 to 24:00.

for the other two phases are of similar trends and neglected here. Three different control strategies including no VAR support, distributed design, and the proposed HVC scheme are plotted. Under a quasi-static operating condition within each minute, the proposed A-HVC algorithm is implemented with each iteration updated every 2 seconds (a total of 30 iterations per minute). Same settings hold for the distributed strategy developed in [10]. This update rate turns out to be sufficient for both the distributed and A-HVC algorithms to achieve satisfactory convergence within a minute interval. For the benchmark case of no VAR support, there are some under- and over-voltage issues due to load and solar variations. Around the noon hours, all three strategies have similarly high voltage mismatch error. This is because the active solar power generation has reached inverter limits during these hours, and accordingly the VAR limits are nearly zero.

To corroborate A-HVC's ability to adapt to varying communication rates, we have considered the worst-case scenario of total communication outage from hour 16:00 to 24:00. The limits of VAR resources would gradually increase from 16:00 on as the solar power generation reduces and the inverters have more capability to control the VAR outputs. Nonetheless, the total link failures during these hours render the distributed control design not responsive to the dynamic operating point at all, resulting in a highest voltage mismatch error from 16:00-18:00 even compared to the benchmark no VAR scenario. This is due to the fact that distributed design would freeze its VAR solution by setting $q_j^{k+1} = q_j^k$ at every node. Meanwhile, the proposed A-HVC design can still effectively minimize the network-wide voltage mismatch error and gracefully maintain a nearly flat voltage profile. Thanks to its flexible adaptivity to communication availability, our A-HVC design enjoys a satisfactory worst-case performance, and significantly outperforms the distributed one under the total communication outage. To sum up, the proposed design can efficiently regulate the voltage level by coordinating network-wide VAR resources. Meanwhile, its cognizant feature to the instantaneous availability of communication links is also attractive, considering the limited deployment of cyber infrastructure in distribution networks. Therefore, the proposed HVC design would facilitate the future engagements in inverter-based VAR resources to improve voltage support by accounting for practical constraints in both physical and cyber layers.

V. Conclusion

This paper has developed a communication-cognizant hybrid voltage control (HVC) algorithm for efficiently coordinating fast VAR resources across distribution networks. Accounting for VAR resource limits, we have cast the problem to minimize a specially designed voltage error objective that can integrate the attractive features of both distributed and local control architectures. The PPD iterations are evoked to dynamically update the VAR solutions with local computations and neighboring bus voltage measurements only. Its convergence conditions are analyzed to determine the choices of step-size. Furthermore, to cope up with varying communication rates arising from limited cyber resources, we have generalized to the asynchronous HVC updates that are robust against random link failures and even total communication outage. We have performed several numerical tests to validate the effectiveness of the proposed HVC design using realistic distribution networks and under dynamic operating conditions.

Future work includes rigorously establishing the convergence conditions for the A-HVC algorithm along with its dynamic performance under practical stochastic system setting. Additionally, it is crucial to study the interactions between fast inverter-based VAR control and other slowly-acting voltage regulation devices, and also to investigate potential cyber-security concerns.

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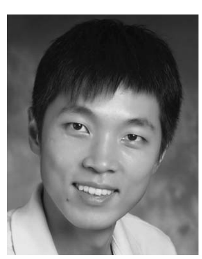
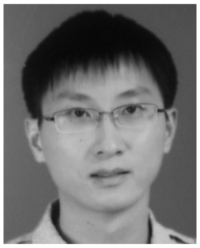


Hao Jan (Max) Liu (M'18) received the B.S. degree from the Missouri University of Science and Technology in 2011, and the M.S. and Ph.D. degrees from the University of Illinois at Urbana-Champaign (UIUC) in 2013 and 2017, all in electrical and computer engineering. He is currently a Research Scientist with the UIUC Information Trust Institute. His research interest lies at the intersection of power systems, control, optimization, and machine learning. He was a recipient of the National Science Foundation East Asia and Pacific Summer Institutes Fellowship in 2015, the 2nd Best Paper Award at the 2016 North American Power Symposium, and the Siebel Energy Scholar Award in 2017.



Wei Shi (M'15) received the B.E. degree in automation and the Ph.D. degree in control science and engineering from the University of Science and Technology of China, Hefei, in 2010 and 2015, respectively. From 2015 to 2016, he was a Post-Doctoral Research Associate with Coordinated Science Laboratory, University of Illinois at Urbana–Champaign, Urbana. He is currently a Post-Doctoral Research Associate with Boston University. His research interests are optimization and its applications in signal processing and control.

Hao Zhu (M'12) received the B.E. degree from Tsinghua University in 2006, and the M.Sc. (minor: Math) and Ph.D. degrees from the University of Minnesota in 2009 and 2012, all in electrical engineering. She is an Assistant Professor of electrical and computer engineering with the University of Texas at Austin. From 2012 to 2013, she was a Post-Doctoral Researcher with the University of Illinois at Urbana-Champaign and as an Assistant Professor of ECE from 2014 to 2017. Her current research interests are investigating and understanding the cyber-physical coupling of power grids for situational awareness and distribution system management, as well as energy data analytics and the broad area of signal processing over networks. She was a recipient the NSF CAREER Award in 2017, the Siebel Energy Institute Seed Grant, and the U.S. AFRL Summer Faculty Fellowship in 2016. She is also the Faculty Advisor and/or co-author of two best papers at the past North American Power Symposium. She is currently a member of the steering committee of the IEEE Smart Grid representing the IEEE Signal Processing Society.